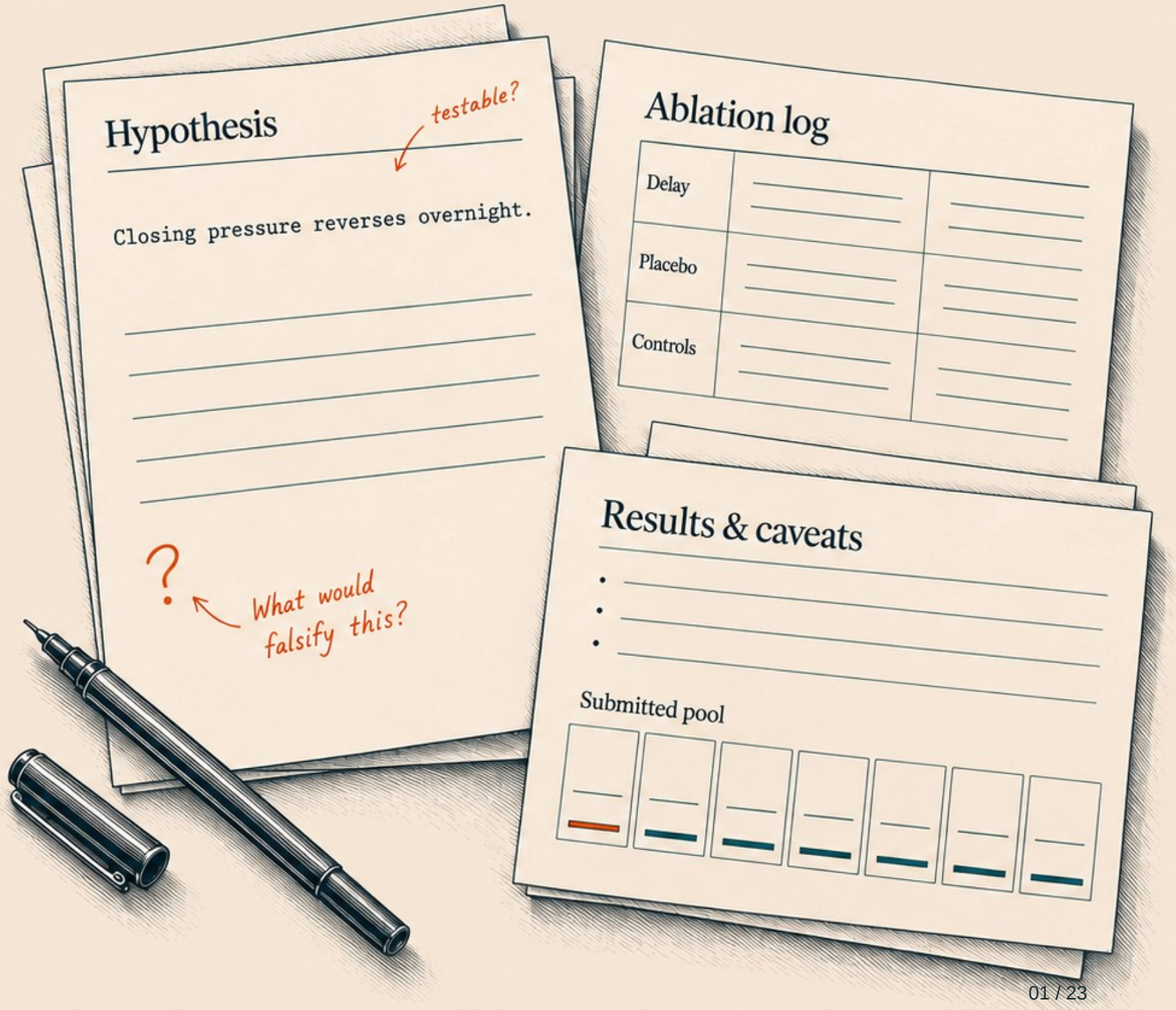
Seven signals. One research discipline.

The hypotheses, ablations and test results behind the submitted pool.



HYPOTHESIS / FALSIFICATION / EVIDENCE

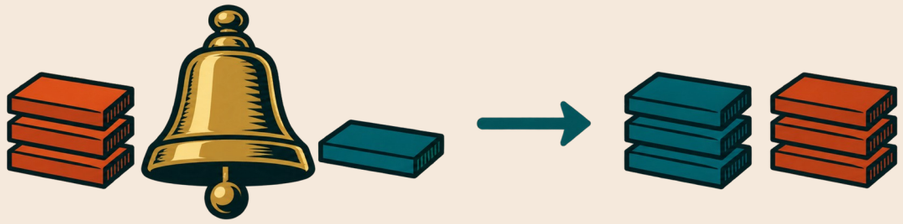
Closing pressure reverses overnight.

THE ECONOMIC IDEA

Index funds, ETFs and benchmarked managers care when they trade. One-sided closing flow pushes the print.

THE PREDICTION

Liquidity providers demand an overnight concession. Two-sided flow returns next morning; the push unwinds.



Closing imbalance Next morning

CONSTRUCTION

```
push = close / vwap_last_hour - 1

alpha = -push
```

Why the last-hour VWAP?

A robust reference price: where the last hour’s volume actually traded.

The test that matters

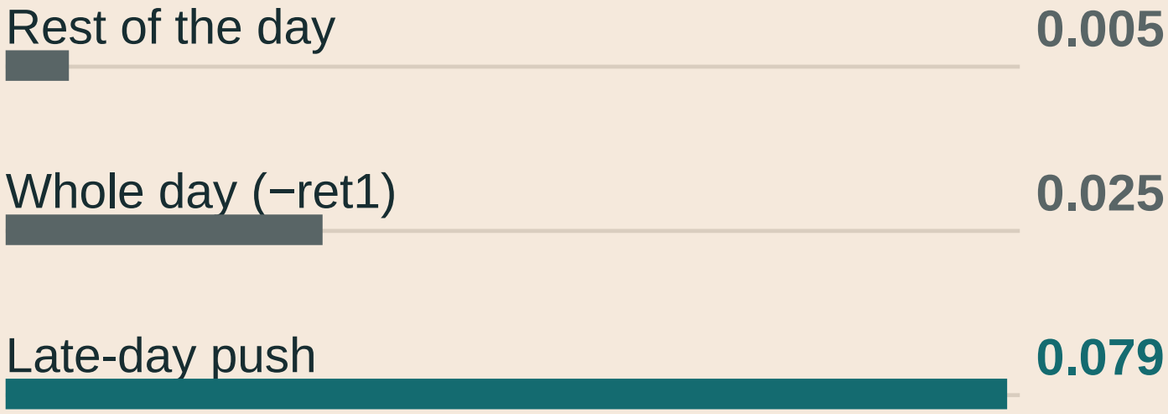
Is this closing-auction pressure, or ordinary short-term reversal?

Bogousslavsky & Muravyev (2023) · schematic illustration; construction shown as pseudocode

The late-day push carries the signal.

```
ret1 ≈ (vwap_lh / prev_close − 1) + (close / vwap_lh − 1)
```

TRADE EACH PART AS REVERSAL



One-day delay: 0.08 → 0.02

Rapid decay fits an overnight unwind. The same delay leaves −ret5 intact.

PnL correlation: 0.17 / 0.05

Versus −ret1 and −ret5; the two reversal baselines correlate 0.65.

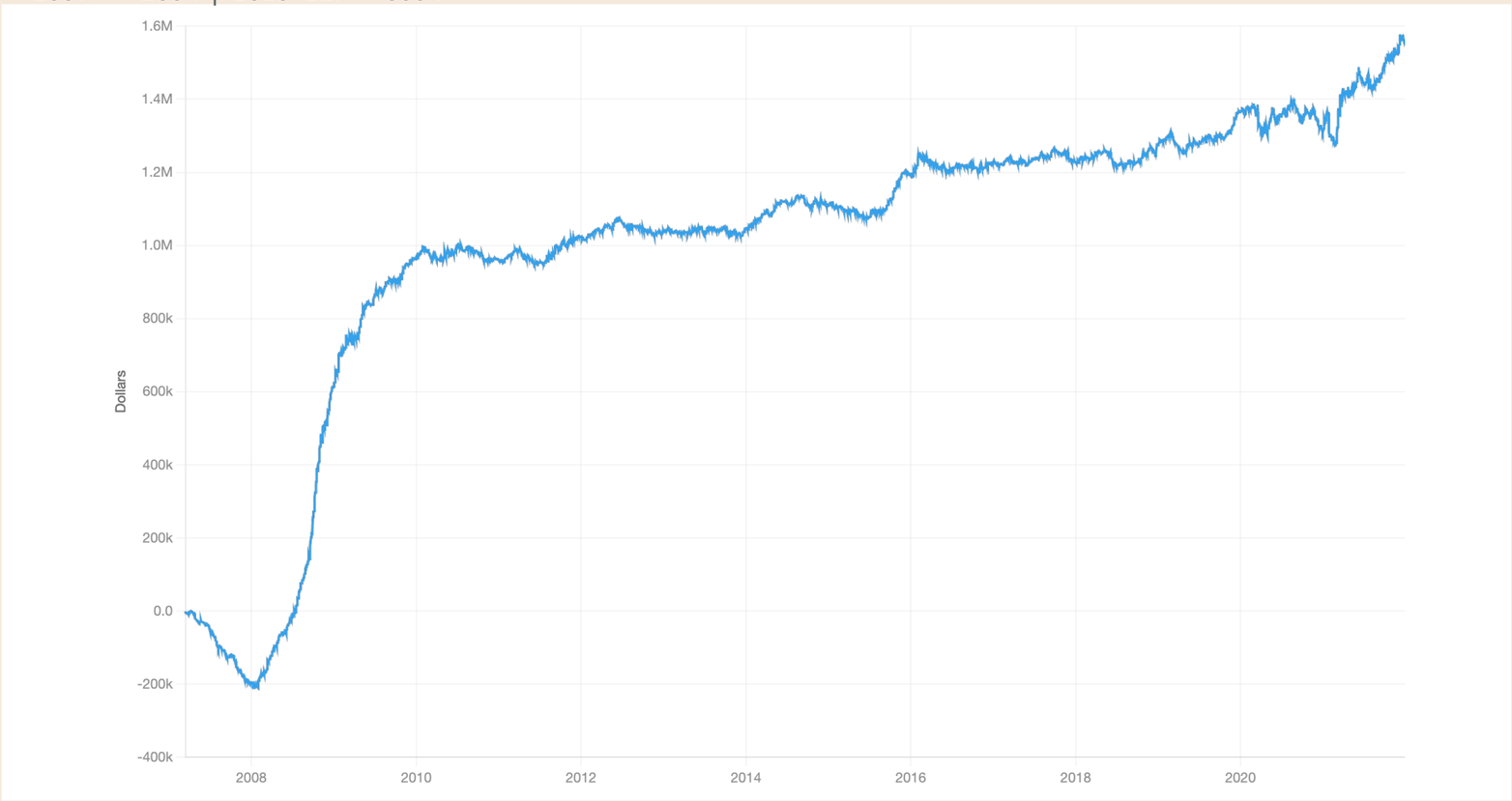
Rejected a higher-IR version

Last-hour close stopped varying after 2011. Its 0.10 slow IR came from 2008–10; VWAP restored coverage.

A strong result. A concentrated crisis gain.

CUMULATIVE PNL / PLATFORM DOLLARS

2007: ≈ −200k | 2010–21: ≈ +600k



≈45% of total PnL came in Q4 2008–mid 2009.

DAILY IR

0.091

t ≈ 5.5 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**1.26**
IR / √TVR	**0.081**
Return (platform)	**0.107**
Max drawdown	**21.7**
Stocks per side	**776 × 773**
Liq / LiqN	**241 / 105**

Above the 25% episode rule.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

The sign flipped. The hypothesis changed.

01 / REJECTED: LOTTERY SKEW

Predicted high skew → low returns. The observed sign ran the other way, and strengthened with controls.

02 / REJECTED: JUMP UNDERREACTION

Predicted weeks of drift and a small-cap effect. Both written predictions failed.

03 / PROPOSED: SPLIT-ORDER FLOW

Parent orders worked over days leave sharp, one-sided minutes. Skew acts as a signed-jump proxy.

CONSTRUCTION

```
rs = ½(skew_first_hour +
       skew_last_hour)
vol = ts_std(ret1, 20)
trd = log ts_mean(trades_lh, 20)
hr = rank(ts_mean(hourly_rank, 5))

alpha = cs_factorneut(
  ts_mean(rs, 5),
  [ret5, vol, log mktcap, trd, hr])
```

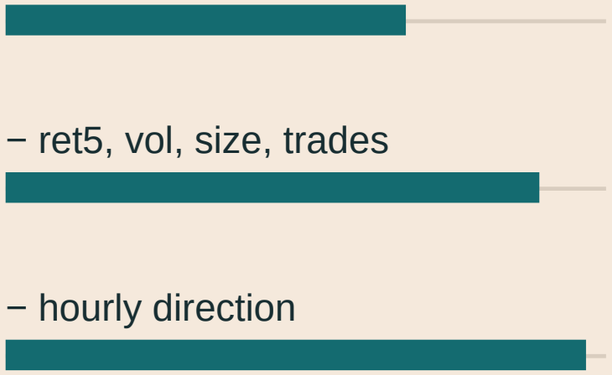


Parent order Child orders

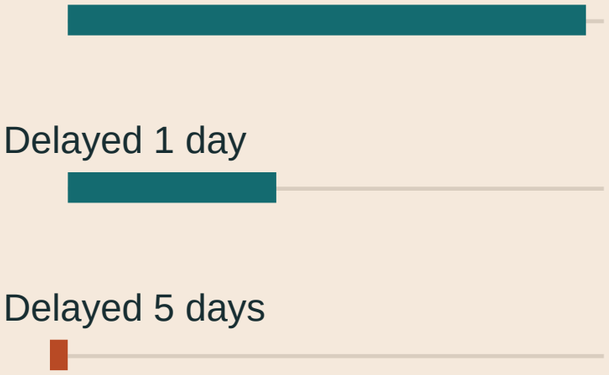
Shape of moves, after removing their direction.

Shape matters. Weeks of drift do not.

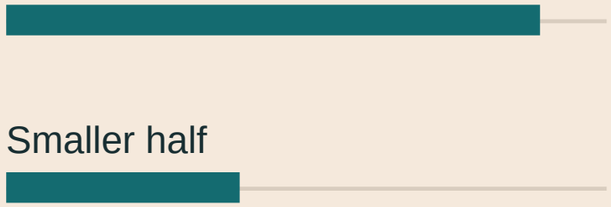
CONTROLS STRENGTHEN IT



DECAY: DAYS, NOT WEEKS



LARGE CAPS CARRY IT



Consistent with institutional order pressure finishing over a few days.

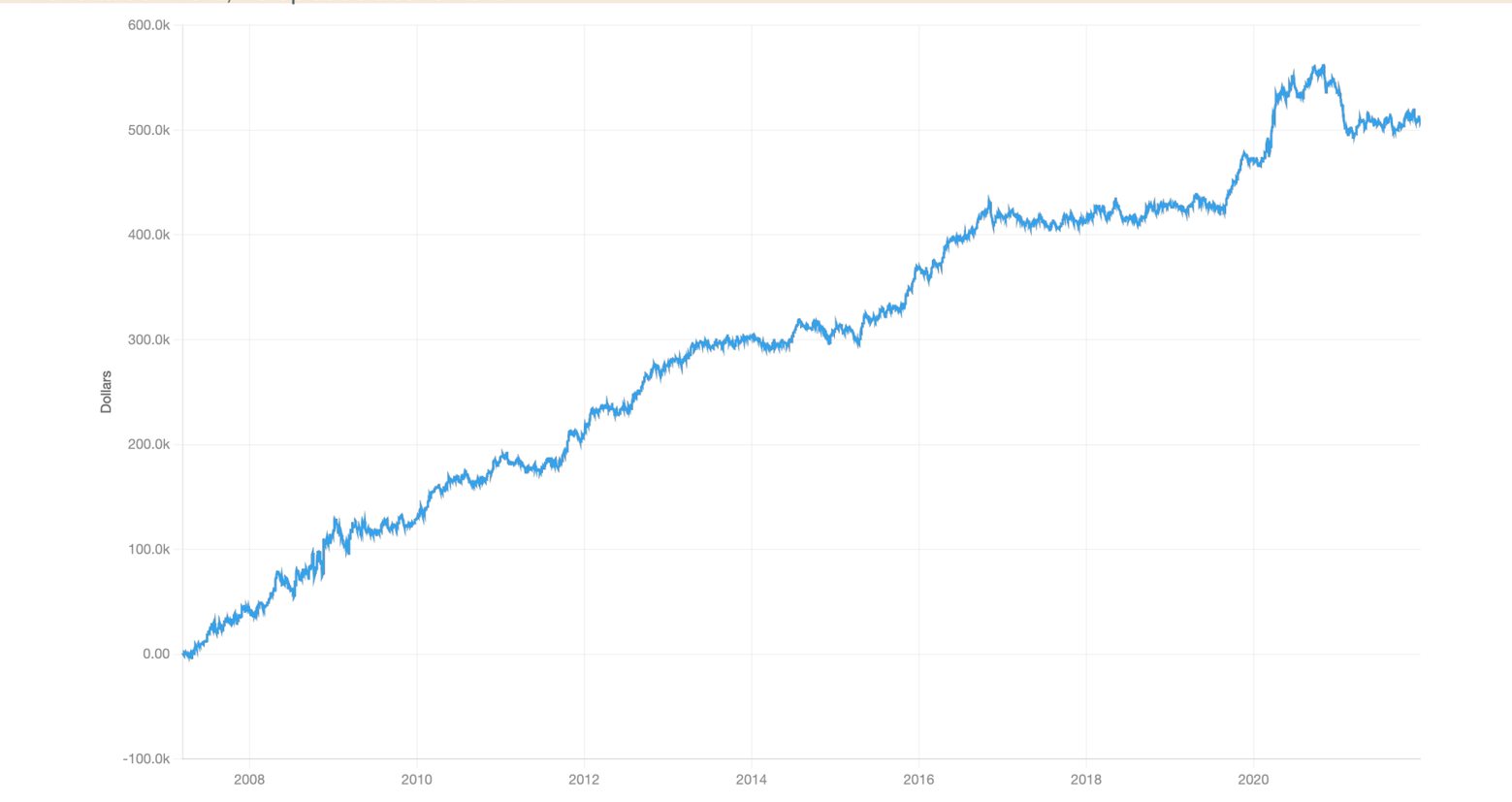
Open issue

PnL is flat from 2017 to mid-2019. Execution changes or crowding are untested explanations.

Steady to 2016. Then a flat stretch.

CUMULATIVE PNL / PLATFORM DOLLARS

2020 adds ≈15%; no episode above 25%.



2017–mid 2019 is the main unresolved flat period.

DAILY IR

0.073

t ≈ 4.4 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.70**
IR / √TVR	**0.088**
Return (platform)	**0.034**
Max drawdown	**7.0**
Stocks per side	**776 × 773**
Liq / LiqN	**572 / 239**

0.003 above the IR bar.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

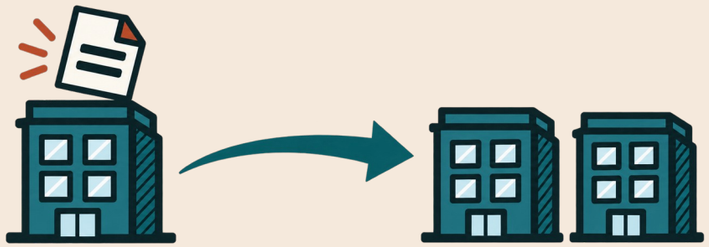
Peer news travels through the trading day.

THE ECONOMIC IDEA

News about related firms reaches the laggard slowly. Trade peers' moves with the stock's own return removed.

TWO PROXIES FOR RELATED FIRMS

A: top co-movers, re-picked annually.
B: cap-weighted industry leaders.
Use daytime returns to isolate the peer news.



Peer news Related firms

CONSTRUCTION

```
day5 = ts_sum(close / open - 1, 5)

# A: top 50 peers, past 252 days
pRaw = peers on raw ret1
pRes = peers on subindustry-net ret1
sA = β × mean(day5 over pRaw)
sB = β × mean(day5 over pRes)
# remove own ret5, ret1; neutralize

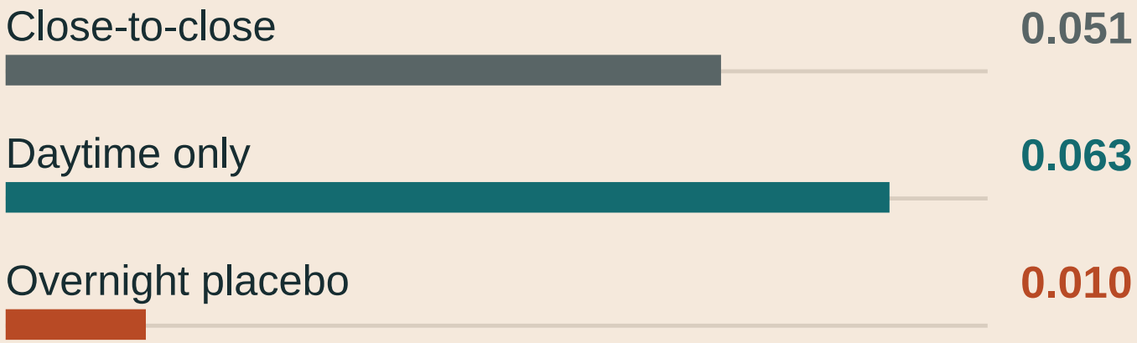
# B: industry leaders
lead = cap-weighted industry ret20
s2 = neut(lead × (2-rank(mktcap)),
          [ret5, ret20])
alpha = zmean(zmean(sA, sB), s2)
```

Equal z-score weights; no tuning.

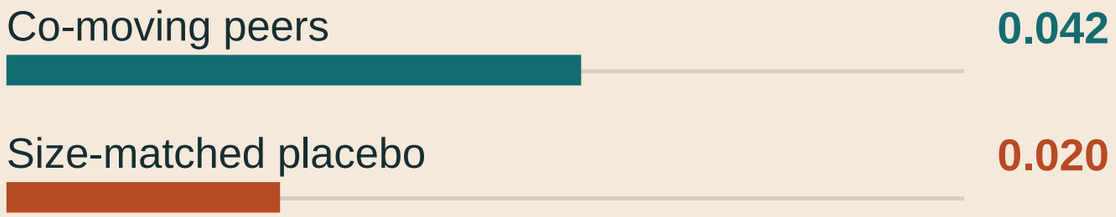
Hou (2007); Cohen & Frazzini (2008); Wang (2025) · peers chosen using past-year data only

The signal survives two placebos.

WHICH HALF OF THE PEER RETURN?



LINKED PEERS OR SHARED FACTORS?



Industry leaders	+ correlation peers	+ second peer set	Daytime peer returns
0.047	**0.071**	**0.077**	**0.091**

Replacement rule met

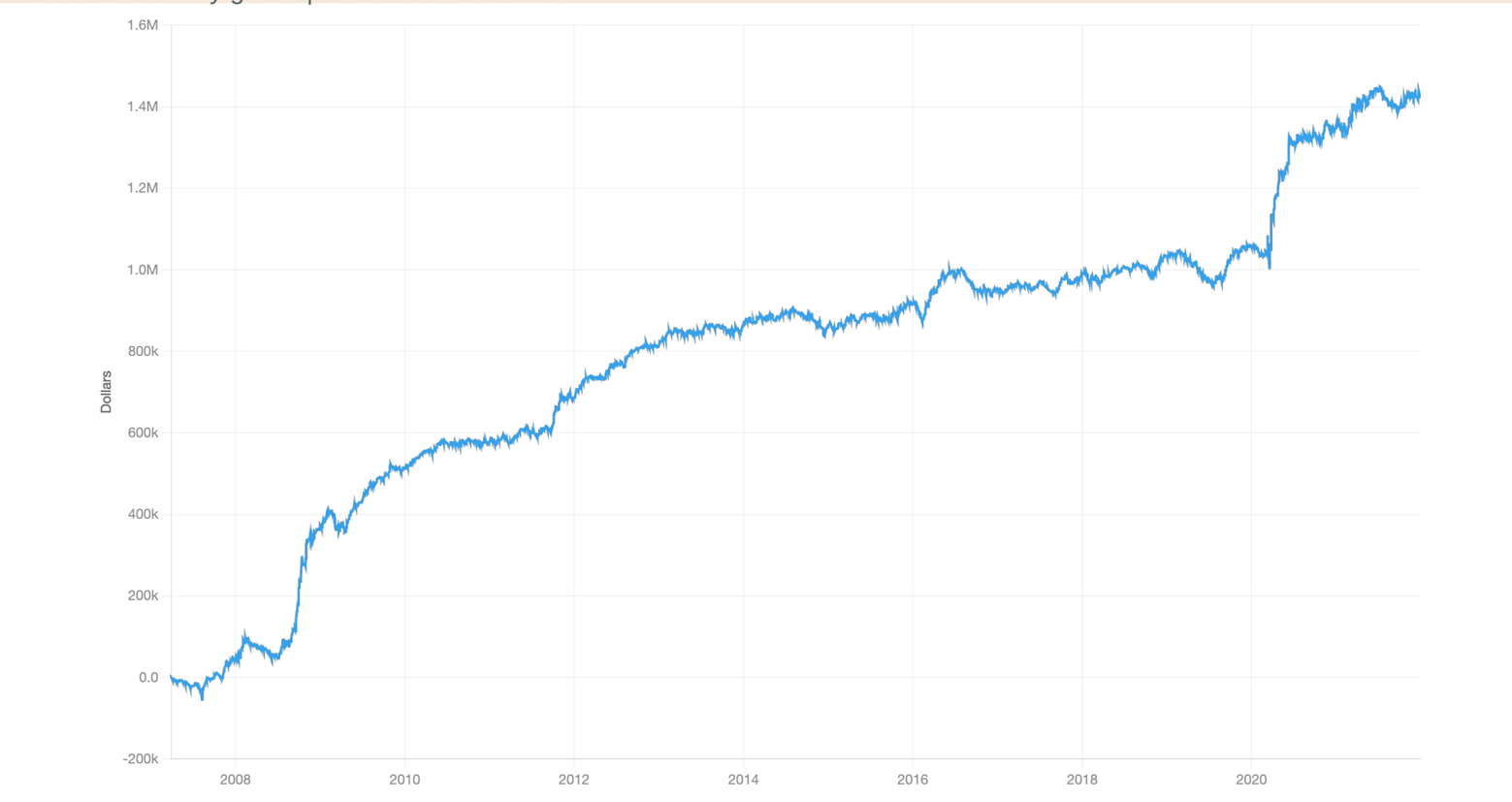
Beat 0.077 IR and 0.094 IR/√TVR. The daytime version achieved 0.091 and 0.109.

Top panels: fast-mode IR · bottom: slow-mode development path · swap rule registered before the replacement run

Calm-year gains, plus crisis rebounds.

CUMULATIVE PNL / PLATFORM DOLLARS

2009–16: steady gains | 2020 rebound: ≈21%



Largest episode ≈21%; nearly flat mid-2016–Feb 2020.

DAILY IR

0.091

t ≈ 5.5 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.69**
IR / √TVR	**0.109**
Return (platform)	**0.098**
Max drawdown	**8.5**
Stocks per side	**785 × 764**
Liq / LiqN	**704 / 332**

Best price-based IR / √TVR.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

Three small effects. Different clocks.

S1 / FIRST-HOUR CONTINUATION

About a week

Split orders recur at the same time of day. Alone: slow IR 0.054.

S2 / MONTH-END REVERSAL

Three days a month

Settlement-driven selling reverses. Alone: 0.039; ≈0.10 per active day.

S3 / RANGE-SHOCK VOL-OF-VOL

About a month

Underperformance after a large-range day. Alone: slow IR 0.031; loses in crashes.

T = last trading day · cv20 = 20-day std / mean

CONSTRUCTION

```
s1 = neut(ts_mean(ret_first_hour, 5),
 [ret5, ret1, vol, log mktcap,
  mean closing push])

s2 = -ret5 as of T-4
# held T-4…T-2; NaN off-window

s3 = zmean(-cv20(log(high/low)),
           -cv20(hourly σ))

alpha = zmean(zmean(s1, s2), s3)
```



Heston, Korajczyk & Sadka (2010); Etula et al. (2020) · weights fixed before the hedge's slow result

A hedge helps. Concentration remains.

PART	FALSIFICATION	RESULT
Month-end	Same trade 10 days earlier	**0.004 vs 0.039**
First hour	Delay for a week	**Keeps ≈60%**
First hour	Last hour instead	**Opposite sign**
Vol-of-vol	Volatility level alone	**0.00**
Vol-of-vol	Drop recent earnings	**Keeps 0.03**
Vol-of-vol	Drop top range day	**0.035 → 0.01**
Vol-of-vol	Delay 20 days	**0.035 → 0.01**

WHY ADD THE THIRD LEG?

45% → 35%

Share of PnL from
Q4 2008–mid 2009

Still above the 25% rule

The hedge loses in crashes and earns in calm years. It improves timing but does not solve concentration.

Honest label: short-horizon underperformance after a non-earnings range shock.

Less crisis exposure, still above the rule.

CUMULATIVE PNL / PLATFORM DOLLARS

The hedge largely removes the March 2020 spike.



≈35% crisis share; 2010–15 is nearly flat.

DAILY IR

0.071

t ≈ 4.3 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.53**
IR / √TVR	**0.097**
Return (platform)	**0.061**
Max drawdown	**8.0**
Stocks per side	**796 × 753**
Liq / LiqN	**630 / 230**

Concentration remains a blocker.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

One event. Two different trades.

BEFORE / DAYS −5 TO −1

Liquidity gets expensive.

Inventory risk rises before the release. Liquidity providers demand larger concessions.

CONSTRUCTION

```
tdu = trading days to next earnings
pre = -ret1 if 1 ≤ tdu ≤ 5
# NaN outside the window
```

~56 stocks per side
Pre-earnings leg

AFTER / DAYS +1 TO +10

The reaction drifts.

Prices continue in the direction of the news. Freeze the reaction, then hold it.

CONSTRUCTION

```
ds = trading days since earnings
R3 = 3-day daytime return / σ60
post = R3 at ds=1, held ds=1…10
```

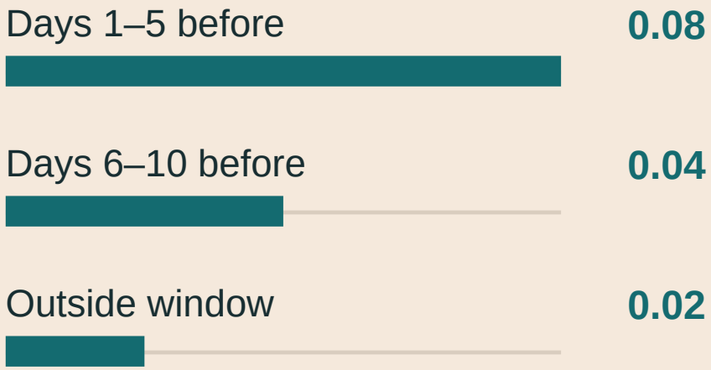


~128 different stocks
Post-earnings leg

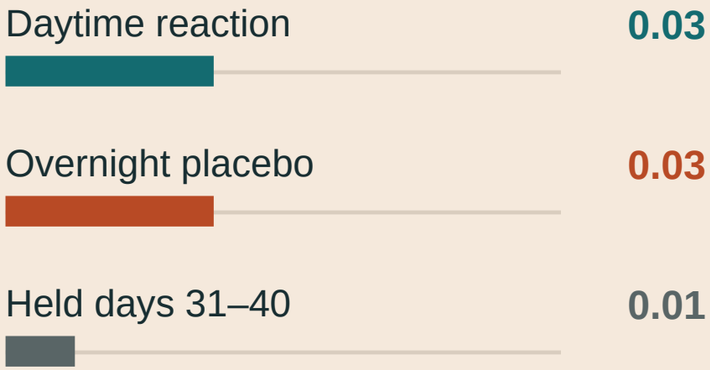
So & Wang (2014); Jiang, Li & Wang (2021) · disjoint windows, equal per-stock z-score weights

A weaker second leg buys breadth.

PRE: DOSE–RESPONSE



POST: A FAILED GATE



Placebo = main.
Relabel as ordinary post-earnings drift.

SLOW RUN	PRE ONLY	FACTOR 5
IR	0.078	**0.070**
Turnover	1.53	**0.77**
IR / √TVR	0.063	**0.080**
Max DD	57.5	**22.2**
Stocks / side	56	**183**
LiqN	3	**22**

IR dips ≈10%; IR/√TVR rises 27%

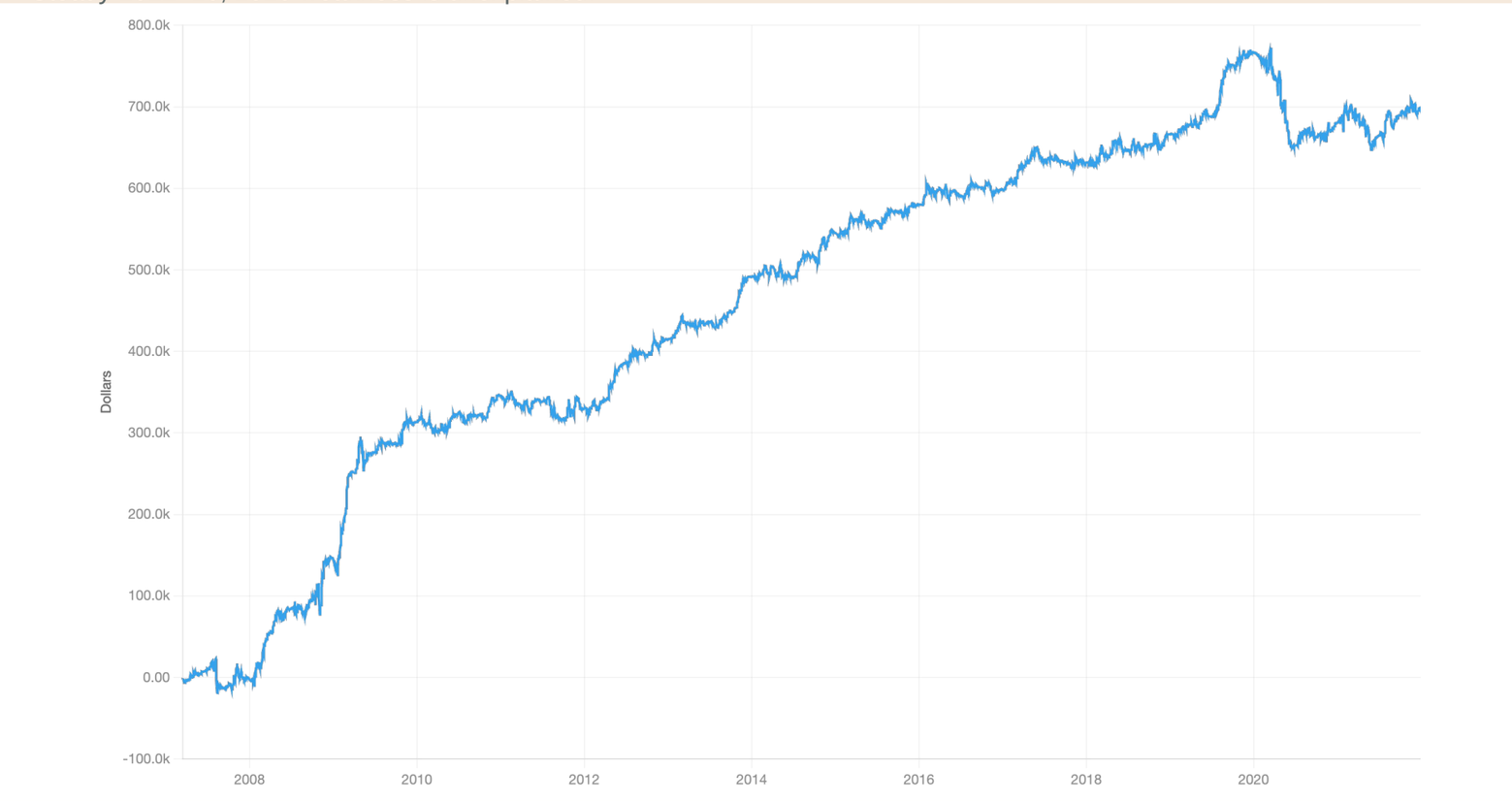
Turnover halves, drawdown falls, the book triples, and capacity grows about sevenfold.

Ablations: fast IR · comparison: slow run · composite run was exploratory: pre-set after-leg trigger of 0.04 was not met

More breadth. An unresolved 2020 loss.

CUMULATIVE PNL / PLATFORM DOLLARS

Steady 2012–16; 2020 weakness is unexplained.



Crisis share ≈29%; 2020 loses ≈17% of total PnL.

DAILY IR

0.070

t ≈ 4.3 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.77**
IR / √TVR	**0.080**
Return (platform)	**0.083**
Max drawdown	**22.2**
Stocks per side	**183 × 184**
Liq / LiqN	**59 / 22**

Exactly on the 0.070 IR bar.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

Six accounting signals. One investment thesis.

The market misreads how firms invest — and how real their earnings are.

LEG / SIGN	ECONOMIC READING
R&D yield (+)	**Expensed R&D hides value**
Operating growth (+)	**Fundamental momentum**
Inventory growth (−)	**Build-ups signal weak demand**
Goodwill / assets (−)	**Acquirers overpay**
Share issuance (−)	**Firms issue when overvalued**
Tax / assets (+)	**Tax income is harder to manage**

CONSTRUCTION

```
s1 = rank(R&D_expense / mktcap)
s2 = rank(operating_income_growth)
s3 = -rank(inventory_growth)
s4 = -rank(goodwill / total_assets)
s5 = -rank(63-day share growth)
s6 = rank(income_tax / total_assets)

alpha = cs_factorneut(
  zmean(s1, …, s6),
  [log mktcap, r250])
```

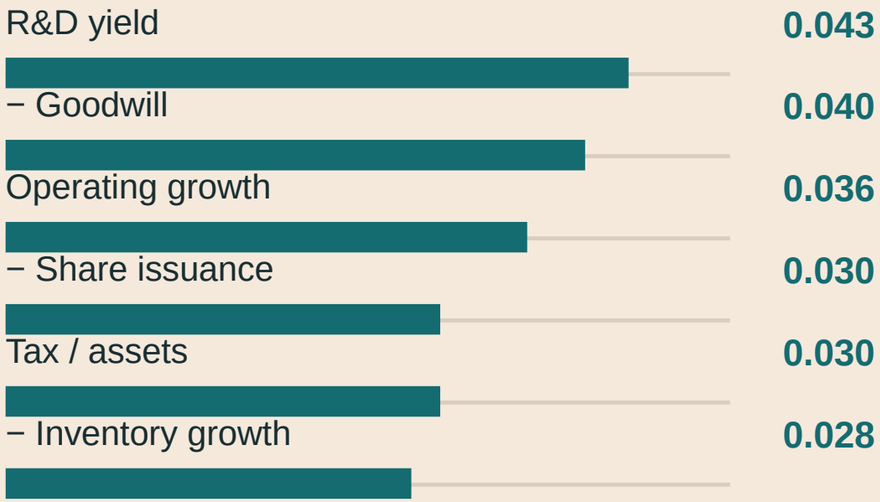


z-scores clipped at ±3; average the legs available for each stock. Remove size and 12-month return.

Chan et al. (2001); Novy-Marx (2015); Thomas & Zhang (2002); Gu & Lev (2011); Pontiff & Woodgate (2008); Lev & Nissim (2004)

Screen by rule. Then test each leg.

EACH LEG ALONE: ABOUT HALF THE BAR



CONFOUND / TIMING TEST	IR	READING
R&D: remove size & 12-m return	0.03	**Survives**
Size placebo within R&D firms	0.01	**Not size**
Inventory: remove op. growth	0.03	**Specific**
Op. growth: remove 12-m return	−0.03	**Price-linked**
Op. growth / inventory: delay 63d	0.00 / 0.01	**Priced in**

First 4-leg composite

0.058

Drop dead leg + controls

0.059

Six legs, pre-set rule

0.082

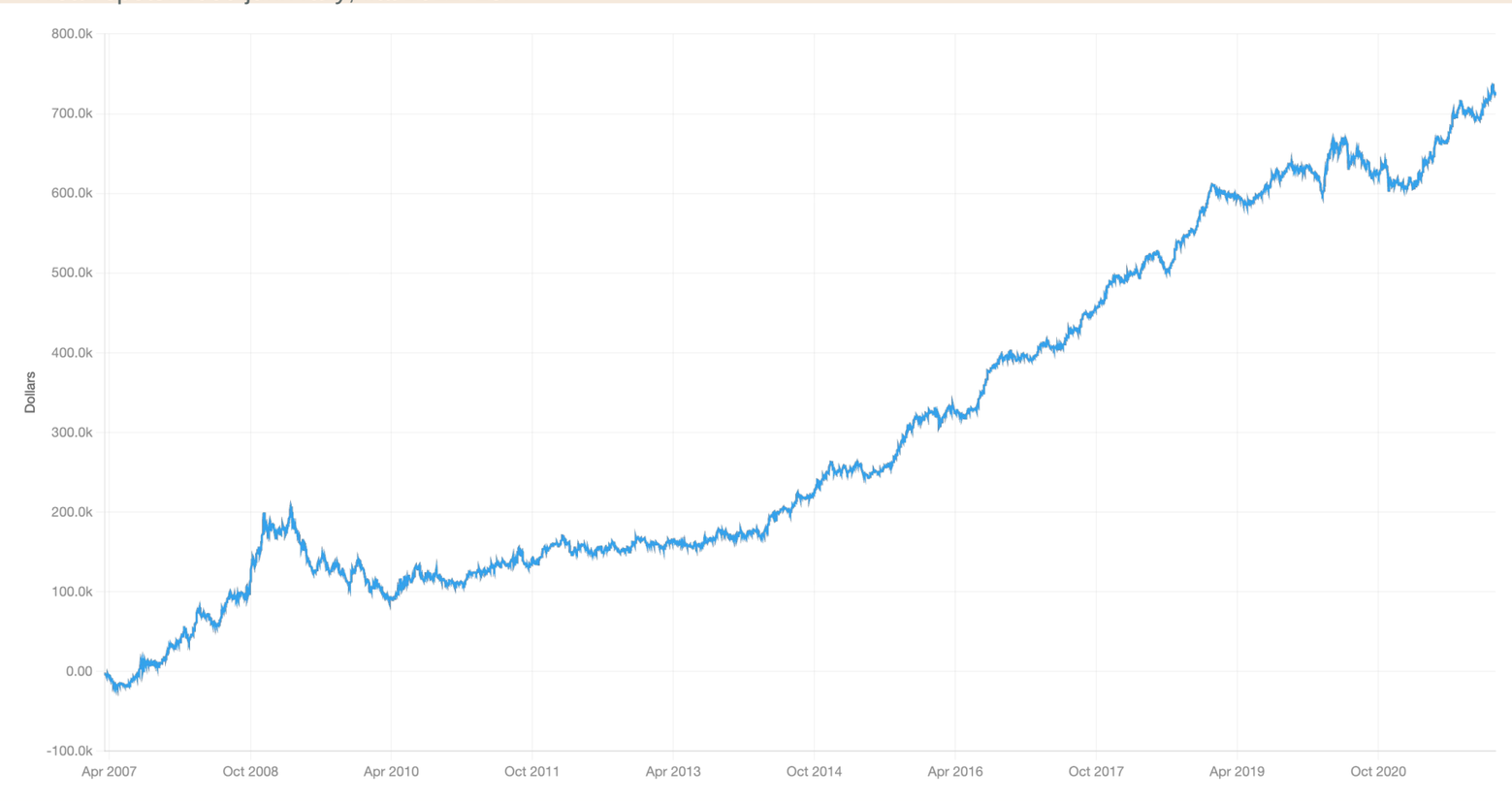
Next test: leave operating growth out. Selection from a screen implies likely out-of-sample decay.

Fast-mode IR for standalone legs and confounds · slow-mode IR for composites · 36 one-line fundamentals screened

The diversifier earns in calm years.

CUMULATIVE PNL / PLATFORM DOLLARS

Weak spots: 2009 junk rally; flat 2011–13.



≈75% of PnL earned in 2010–19; crisis share ≈4%.

DAILY IR

0.082

t ≈ 5.0 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.054**
IR / √TVR	**0.354**
Return (platform)	**0.049**
Max drawdown	**11.9**
Stocks per side	**651 × 631**
Liq / LiqN	**694 / 379**

Lowest turnover in the pool.

2007–21 · original platform PnL image · episode shares are approximate readings from the chart

Over-expansion has two channels.

CAPITAL / BALANCE-SHEET BLOAT

Net operating assets accumulate when accounting earnings outrun free cash flow. Short high NOA.

LABOR / HIRING

Fast hiring predicts low returns. Here headcount is inferred; much of the leg is earnings growth the price has not confirmed.



Capital



Labor

CONSTRUCTION

```
r250 = close / delay(close,250) - 1
op_assets = total_assets - cash
op_liabs = total_liabs - total_debt
noa = (op_assets-op_liabs) / assets
emp = ttm_eps × shsout / ni_per_emp

sA = -rank(noa)
sH = neut(-rank(emp/emp_1y_ago-1),
          [log mktcap, r250])
alpha = cs_factorneut(
  zmean(sA, sH), [log mktcap, r250])
```

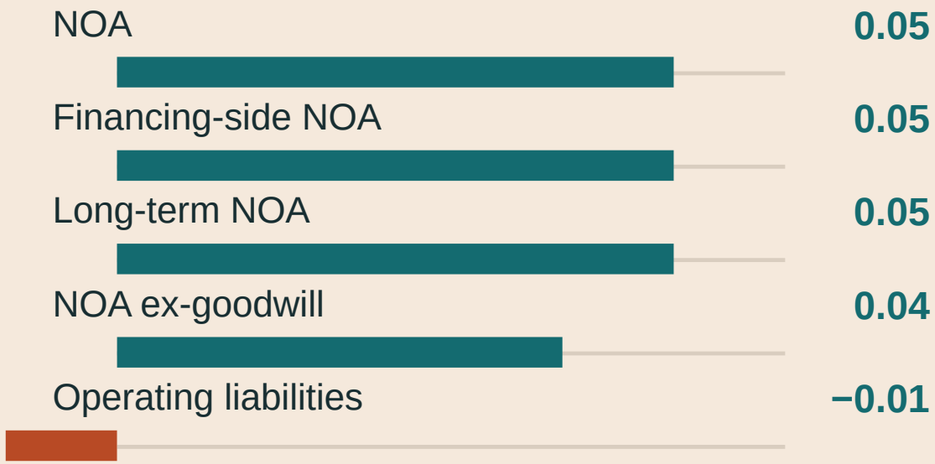
Leg correlation: −0.17

Use NOA alone when hiring is missing. Clip each leg’s z-score at ±3.

Hirshleifer, Hou, Teoh & Zhang (2004); Belo, Lin & Bazdresch (2014) · inferred headcount, profitable firms only

NOA is robust. Hiring is a mixed proxy.

NOA WORKS AS A NET FIGURE



HIRING-LEG VERSION	IR	SHARE
As built	0.07	—
Earnings piece alone	0.06	**≈85%**
Earnings growth removed	0.04	**≈57% left**
NOA level / change removed	0.061	**≈87%**
Factor 6 growth legs removed	0.05	**≈71%**

NOA holds in slow mode (0.05), after controls (0.05), and at a 63-day delay (0.06).

Honest label: hiring / price-unconfirmed earnings growth.

NOA alone	Hiring alone	Both, equal weight
0.050	**0.069**	**0.074**

Pre-run estimate: 0.08–0.09. The result came in below it; the hiring leg covers profitable firms only.

Top panels: fast-mode IR · bottom: slow-mode combination · primary NOA version and equal weights fixed before the composite

Steady growth. Almost no crisis share.

CUMULATIVE PNL / PLATFORM DOLLARS

Q4 2008 dip recovered; weak 2012–14; flat since mid-2020.



≈71% of PnL earned in 2010–19; crisis share ≈1%.

DAILY IR

0.074

t ≈ 4.5 · t ≈ IR × √days

SLOW-RUN METRIC	VALUE
Turnover	**0.056**
IR / √TVR	**0.313**
Return (platform)	**0.052**
Max drawdown	**12.8**
Stocks per side	**475 × 458**
Liq / LiqN	**503 / 309**

Pool correlations ≤0.14.

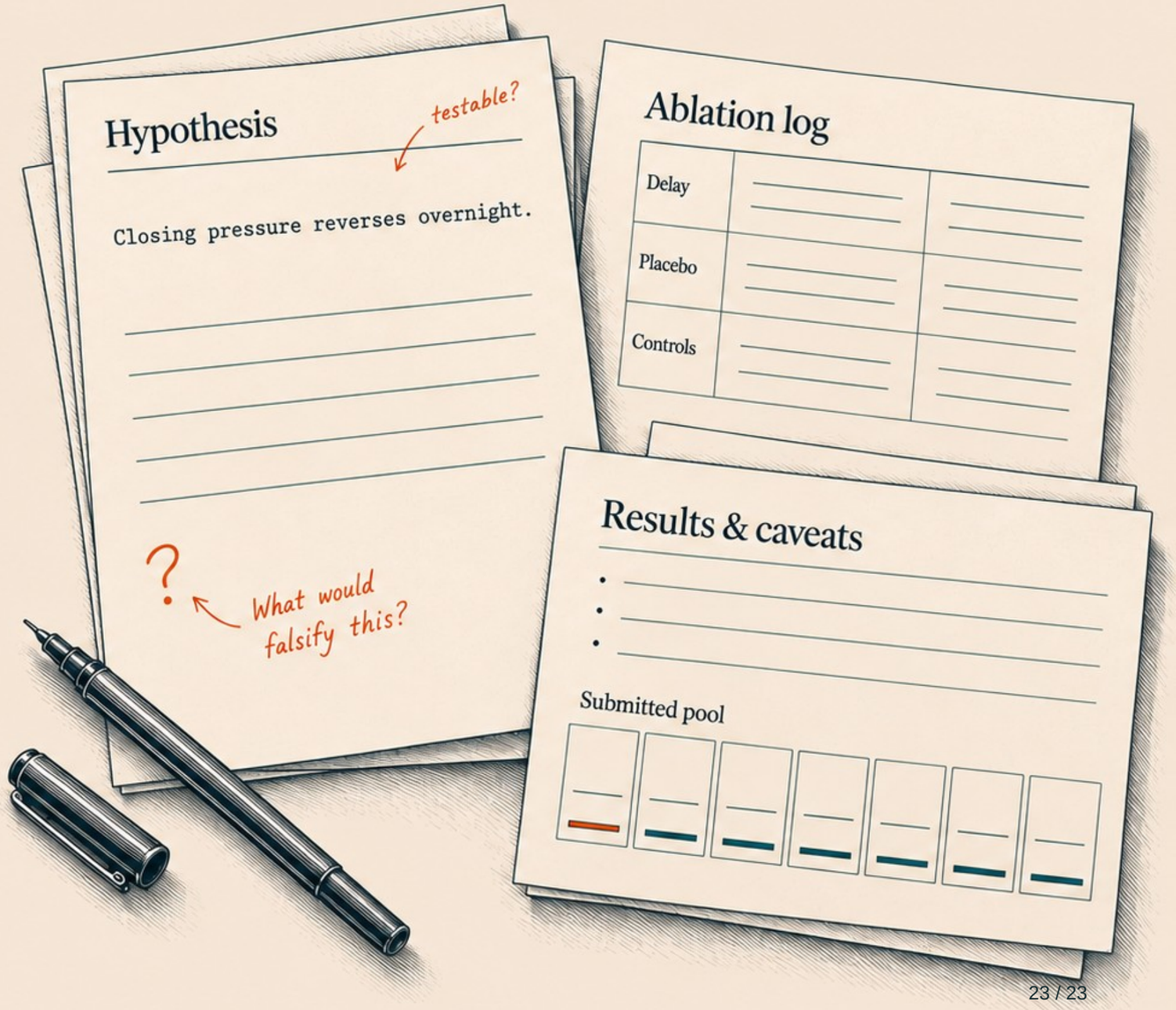
2007–21 · original platform PnL image · episode shares are approximate readings from the chart

Thank you.

The evidence.

The limitations.

The next tests.



SEVEN SIGNALS. ONE RESEARCH DISCIPLINE.